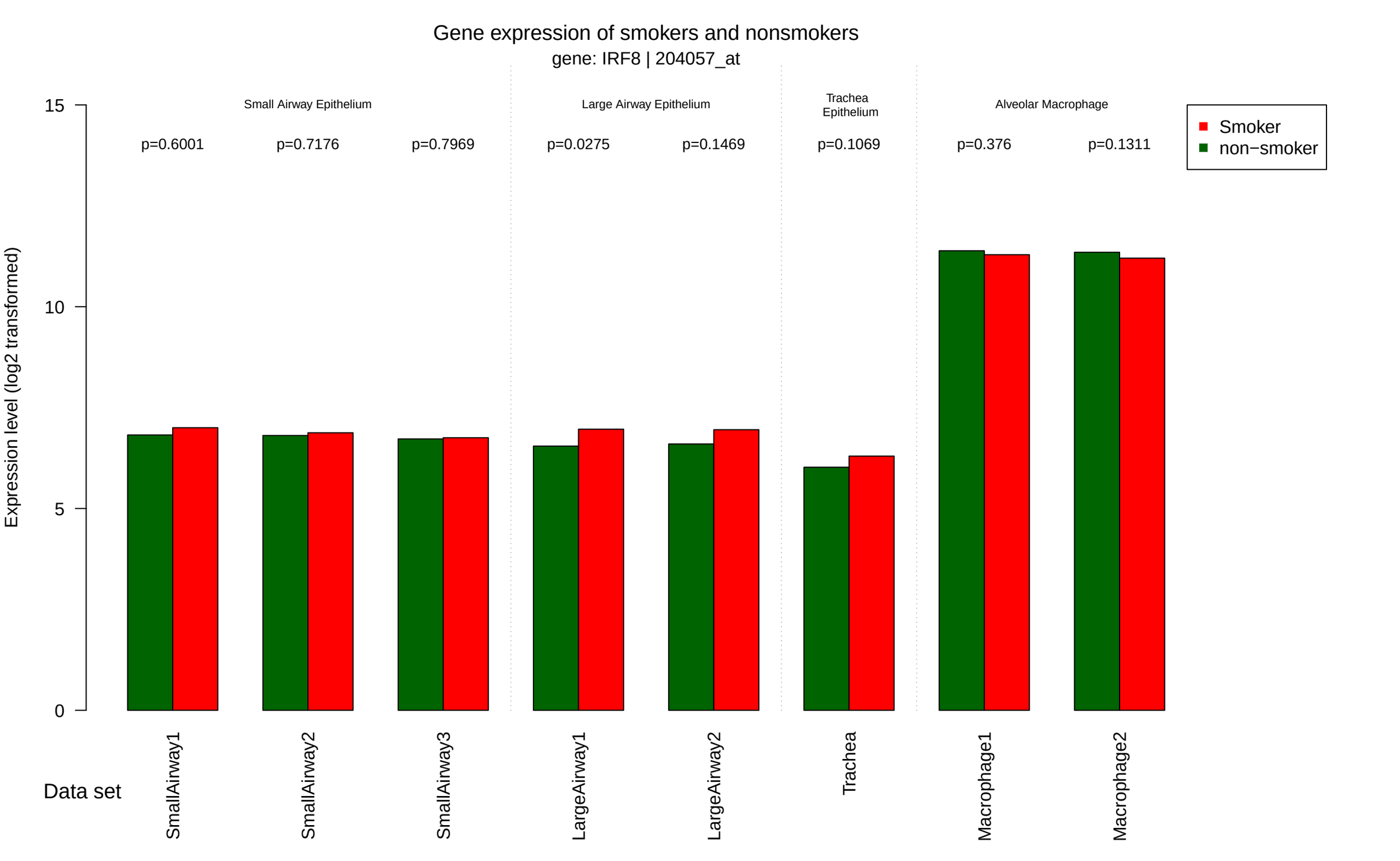
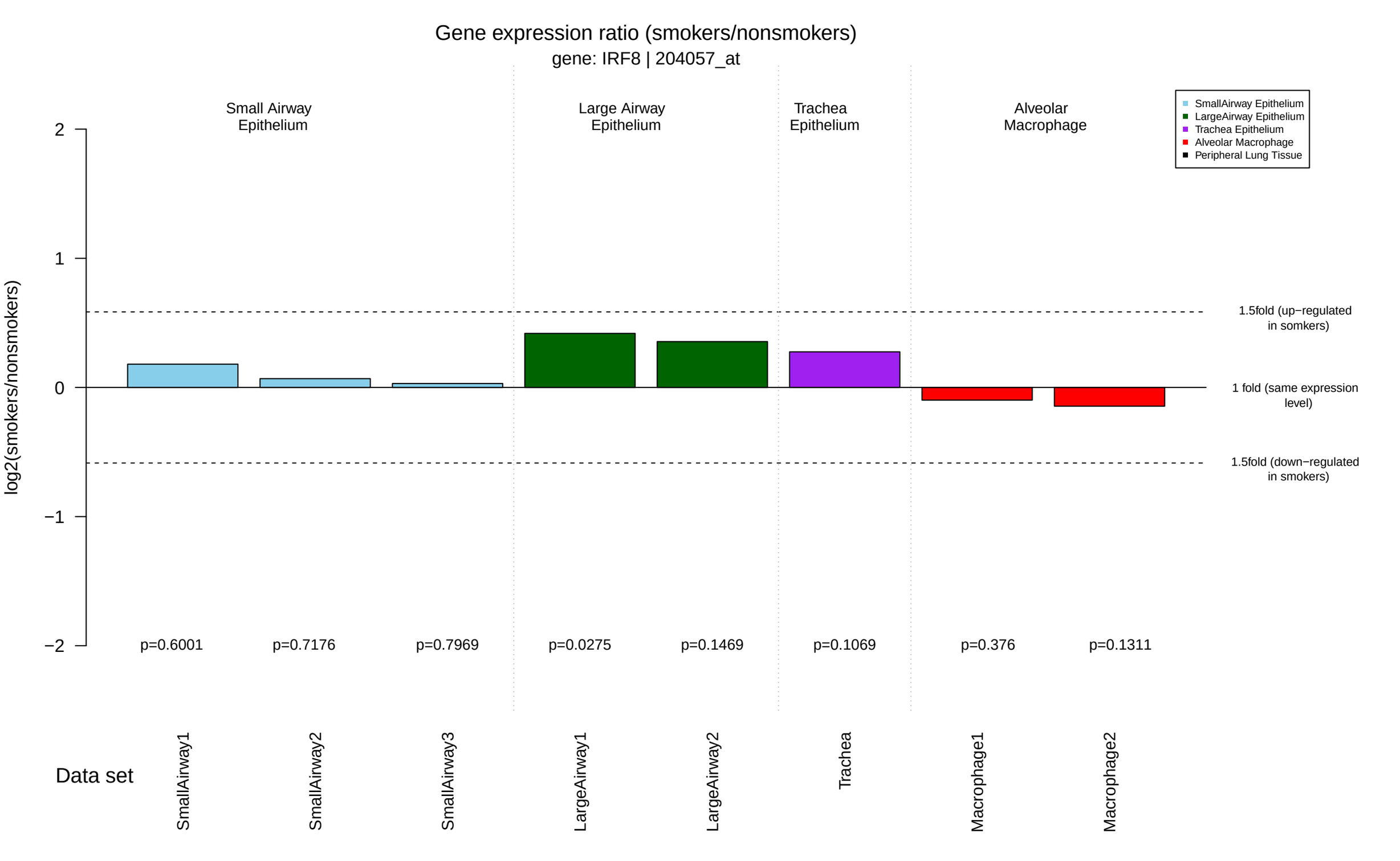
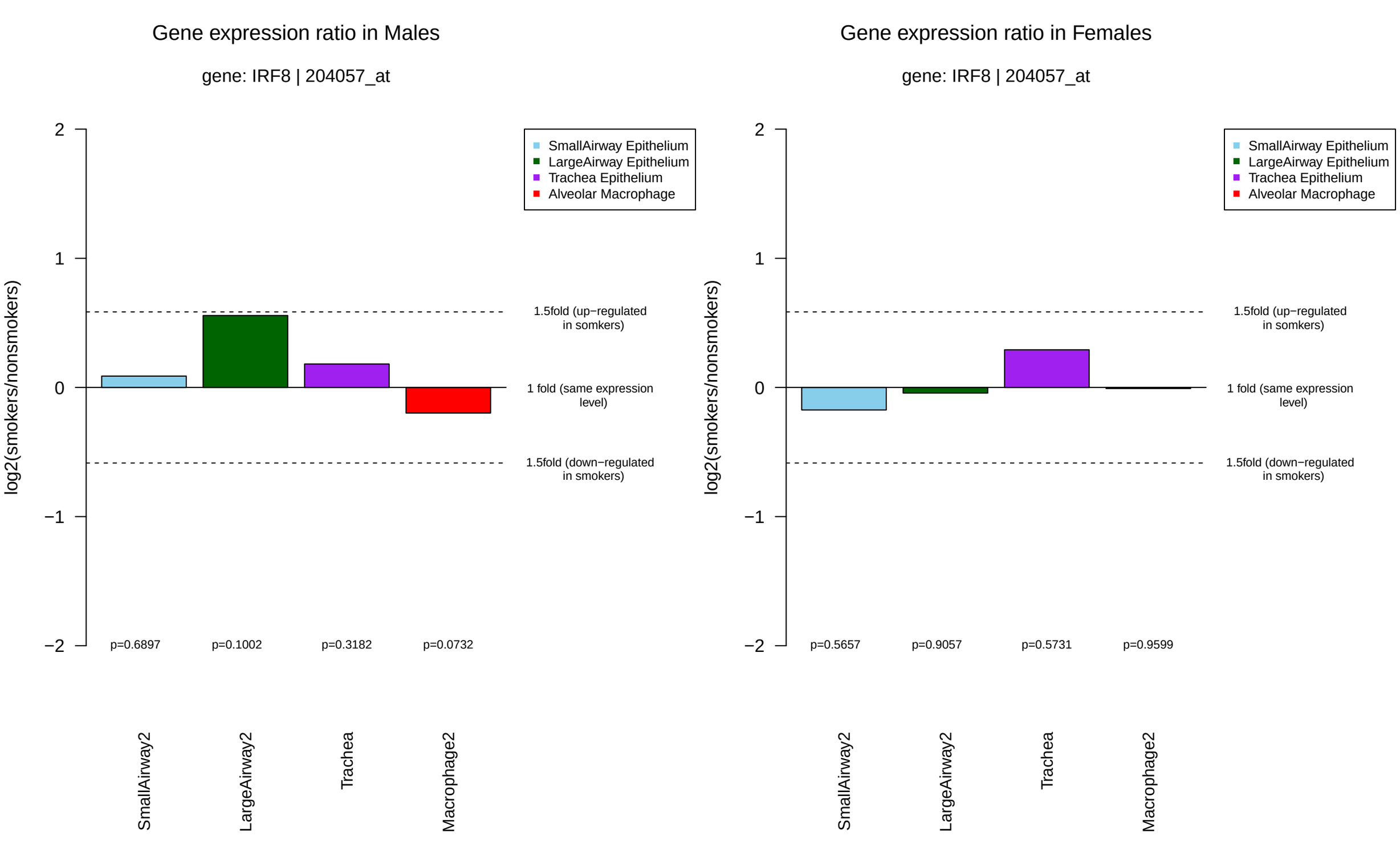
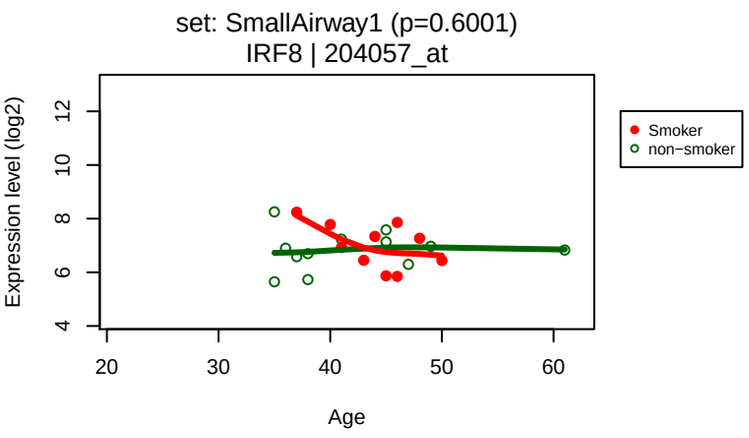
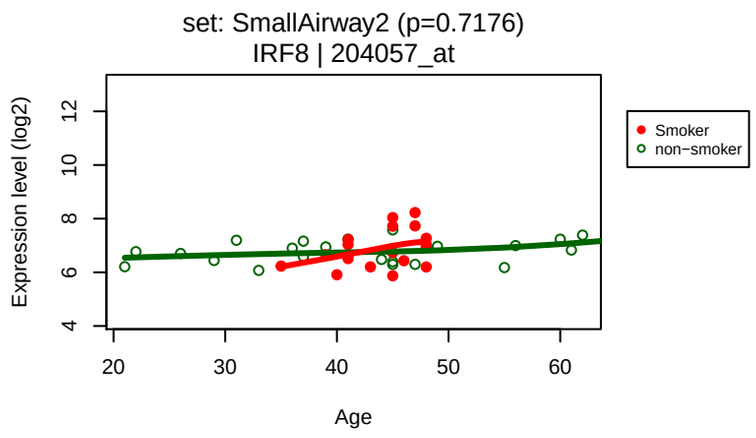
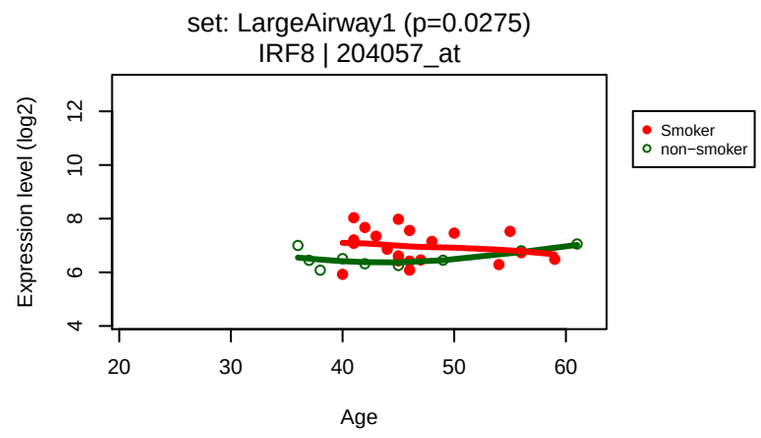

Gene expression of smokers and nonsmokers
gene: IRF8 | 204057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6001
p=0.7176
p=0.7969
p=0.0275
p=0.1469
p=0.1069
p=0.376
p=0.1311
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

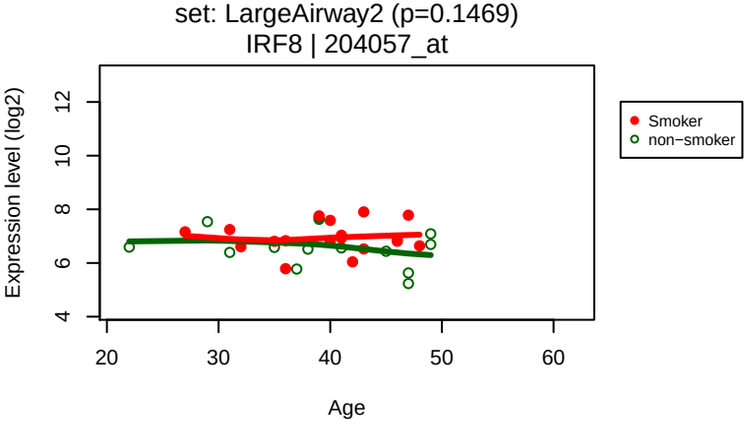

Gene expression ratio (smokers/nonsmokers)
gene: IRF8 | 204057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6001
p=0.7176
p=0.7969
p=0.0275
p=0.1469
p=0.1069
p=0.376
p=0.1311
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

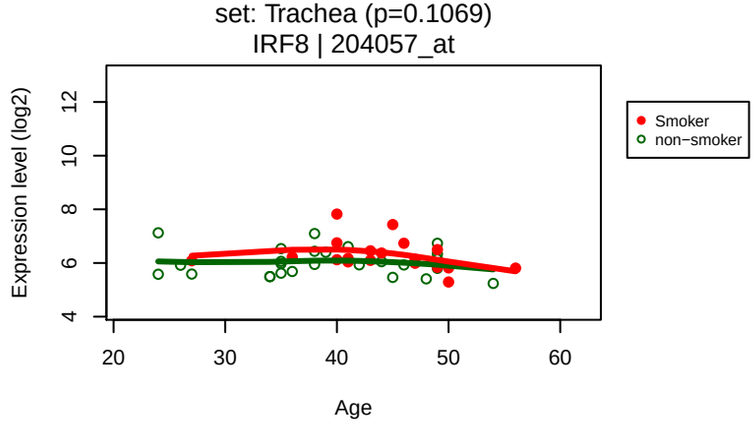

Gene expression ratio in Males
gene: IRF8 | 204057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6897
p=0.1002
p=0.3182
p=0.0732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRF8 | 204057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5657
p=0.9057
p=0.5731
p=0.9599
SmallAirway2
LargeAirway2
Trachea
Macrophage2

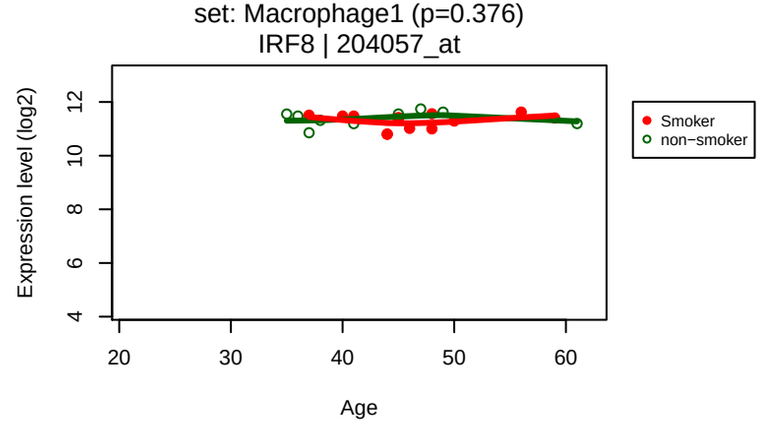

set: SmallAirway1 (p=0.6001)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker

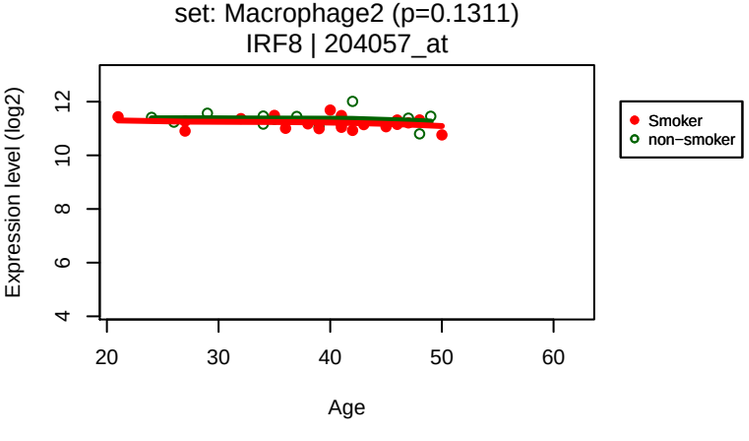

set: SmallAirway2 (p=0.7176)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker

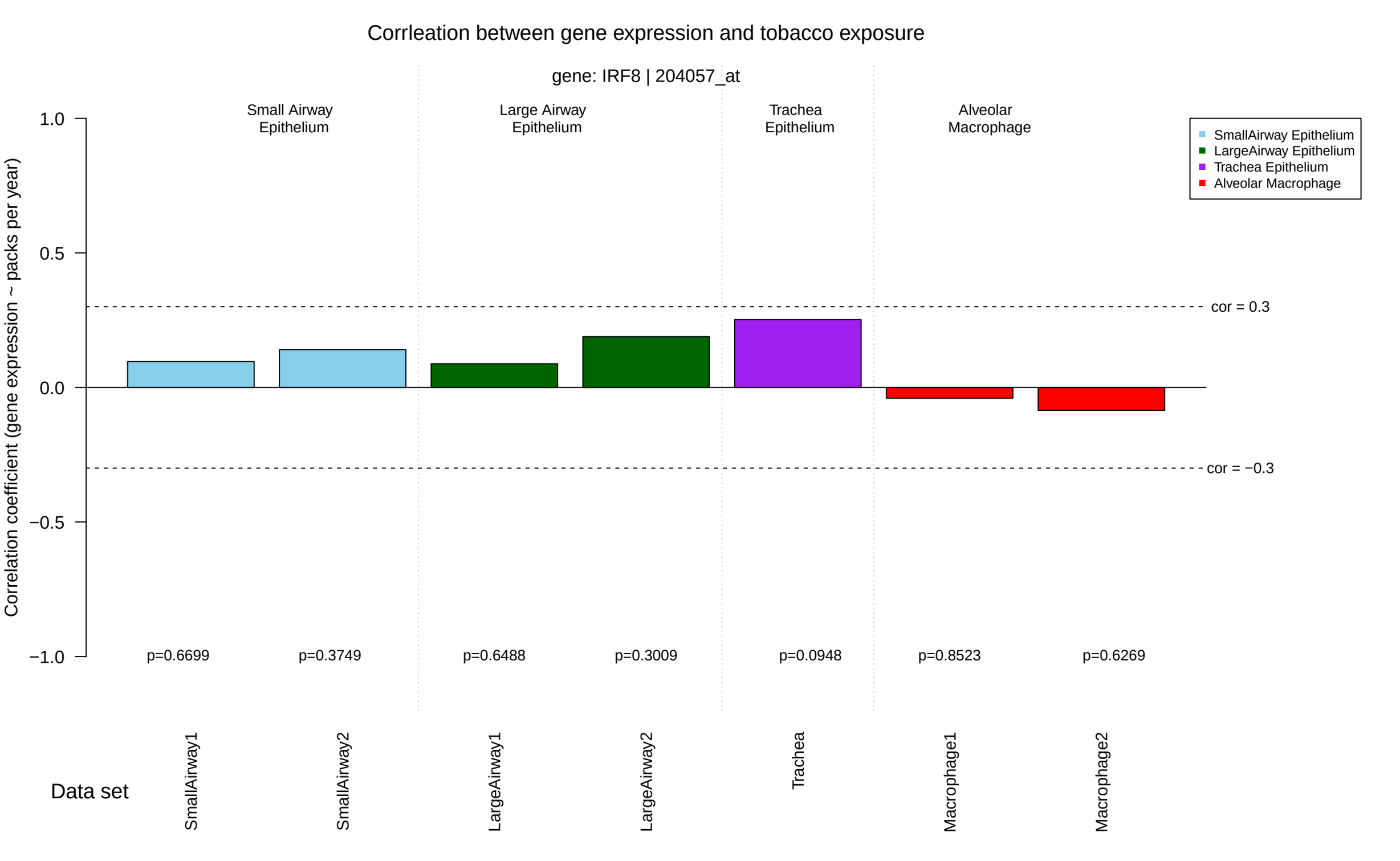

set: LargeAirway1 (p=0.0275)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1469)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1069)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.376)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1311)
IRF8 | 204057_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRF8 | 204057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6699
p=0.3749
p=0.6488
p=0.3009
p=0.0948
p=0.8523
p=0.6269
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2